

SE2030 – Software Engineering

Agile & Scrum – Practice Quiz

60 Questions | Lecture 3: Agile (Modern Software Development Methodologies)

Instructions

This practice quiz covers the full Lecture 3 content: Traditional vs. Modern methodologies, Why Agile, the Agile Manifesto (Values & Principles), Agile methodologies and practices, the Scrum framework (roles, artifacts, activities), writing user stories, and the sprint burn down chart / hospital appointment system examples.

Section A (Questions 1–40): Single-Answer Multiple Choice. Choose the ONE best answer (A, B, C, or D) for each question.

Section B (Questions 41–60): Multiple Select. Each question may have 2 to 4 correct answers out of 5 options (A–E). Select ALL options that apply — partial answers are considered incomplete.

Try to answer every question before checking the Answer Key at the end of this document. Good luck!

SECTION A — Single-Answer Multiple Choice (Q1–40)

Choose the ONE best answer for each question.

1. In a traditional software development model, what typically happens if a customer requests a major requirement change after 6 months into a 12-month project?

- A) The change is implemented instantly with no cost impact
- B) Rework is required, costs increase, and delays can occur
- C) The project automatically switches to an Agile approach
- D) The change is added to the next sprint backlog

2. Which of the following best describes the Agile approach to requirement changes compared to the traditional approach?

- A) Agile discourages any changes once a sprint has started
- B) Agile treats changing requirements as an obstacle to avoid entirely
- C) Agile requires a full 12-month plan before development begins
- D) Agile embraces changing requirements and adapts more effectively than traditional models

3. According to the lecture, which characteristic is associated with the “Traditional Way” of software development?

- A) Lack of transparency
- B) Highest value delivery
- C) High quality within the shortest time
- D) Frequent short iterations

4. Which of these is listed as a benefit of the “Modern Way” (Agile) compared to Traditional development?

- A) Less or no iterations
- B) High cost
- C) Highest value, high quality, shortest time
- D) Changes are not acceptable

5. A customer originally requested a simple online shopping system, but after 6 months realizes they also need mobile support and online payments. In a traditional development approach, what is the most likely consequence?

- A) These features are added for free since planning was flexible
- B) These changes are costly and difficult to implement because development is already underway
- C) The system automatically detects the new requirement
- D) Nothing changes since traditional models are always flexible

6. Which of the following is NOT one of the challenges of traditional software development mentioned in the lecture?

- A) Errors may be discovered late
- B) Customers may not see the system until the end of the project
- C) Teams may build features that users no longer need
- D) Software is delivered in small, frequent increments

7. How does Agile help address the problem of “requirements may change during development”?

- A) By adapting to changing requirements through iterative delivery
- B) By avoiding customer contact until the final release
- C) By locking requirements at the start and never changing them
- D) By extending the project timeline indefinitely

8. The Agile Manifesto was introduced to the software industry around which year, as mentioned in the lecture?

- A) 1995
- B) 1980
- C) 2001
- D) 2010

9. What is the main difference between “Values” and “Principles” in the Agile Manifesto, as explained in the lecture?

- A) Values are specific coding techniques, while principles are business goals
- B) Values represent the core philosophy, while principles guide how values are applied in practice
- C) Values apply only to Scrum, while principles apply only to Kanban
- D) There is no difference; the terms are interchangeable

10. How many key Values and Principles does the Agile Manifesto define, according to the lecture?

- A) 3 Values and 10 Principles
- B) 5 Values and 15 Principles
- C) 4 Values and 10 Principles
- D) 4 Values and 12 Principles

11. Which Agile value emphasizes team collaboration and including customers in the process?

- A) Individuals and Interactions
- B) Working Software
- C) Contract Negotiation
- D) Following a Plan

12. In the Agile Manifesto's comparison table, “Working Software” in Agile is valued over which traditional practice?

- A) Process and Tools
- B) Contract Negotiation
- C) Comprehensive Documentation
- D) Following a Plan

13. Which traditional practice does “Responding to Change” replace in the Agile value comparison?

- A) Comprehensive Documentation
- B) Following a Plan

- C) Contract Negotiation
- D) Process and Tools

14. Which Agile methodology organizes work into short, time-boxed iterations known as Sprints?

- A) Kanban
- B) Extreme Programming
- C) Lean Software Development
- D) Scrum

15. Which Agile methodology is primarily represented using a visual board to manage and track work?

- A) Kanban
- B) Scrum
- C) Lean Software Development
- D) Extreme Programming

16. Extreme Programming (XP) primarily focuses on which of the following?

- A) Eliminating waste and maximizing value
- B) Visual tracking of tasks on a board
- C) Engineering practices and code quality
- D) Defining a Sprint Goal

17. Which methodology's main focus is “eliminating waste and maximizing value”?

- A) Scrum
- B) Lean Software Development
- C) Extreme Programming
- D) Kanban

18. According to the lecture, roughly how long does a typical Scrum sprint last?

- A) 6 to 12 months
- B) A single day
- C) Exactly one year
- D) One to four weeks

19. Which statement best distinguishes Agile “methodologies” from Agile “practices”?

- A) Methodologies provide the overall structure for managing a project, while practices are specific implementation techniques
- B) Methodologies are specific coding techniques; practices define the overall project structure
- C) Methodologies and practices mean exactly the same thing
- D) Practices are only used in Scrum, while methodologies are used in all frameworks

20. In Test-Driven Development (TDD), what is the correct sequence of actions?

- A) Write the code first, then write tests, then refactor
- B) Deploy the software, then write tests
- C) Write the tests before writing the code
- D) Skip testing entirely to save time

21. What is a key benefit of Pair Programming as described in the lecture?

- A) It eliminates the need for any testing
- B) Two developers work together, helping catch bugs and improve code quality early
- C) It replaces the need for a Scrum Master
- D) It is only used in the Waterfall model

22. Behavior-Driven Development (BDD) primarily focuses on defining system behavior using:

- A) Complex technical algorithms only understandable by developers
- B) Database schema diagrams
- C) Sprint burndown metrics
- D) Business-focused scenarios

23. Which of the following best defines Scrum?

- A) An Agile framework that organizes work into Sprints and emphasizes team collaboration
- B) A programming language used for Agile development
- C) A type of documentation used only in Waterfall projects
- D) A single developer working independently without a team

24. Who is primarily responsible for representing the customer and prioritizing requirements in a Scrum team?

- A) Scrum Master
- B) Developers
- C) Project Manager
- D) Product Owner

25. Which Scrum role is described as a “service leader” who supports the team rather than assigning tasks?

- A) Product Owner
- B) Scrum Master
- C) Developer
- D) Business Analyst

26. What is a common misconception about the Scrum Master's role, as clarified in the lecture?

- A) That the Scrum Master writes all the code
- B) That the Scrum Master only works with customers
- C) That the Scrum Master is the same as a traditional project manager or team leader
- D) That the Scrum Master owns the product backlog

27. Which role is primarily responsible for designing, developing, testing, and delivering a working product increment each sprint?

- A) Developers
- B) Product Owner
- C) Scrum Master
- D) Business Analyst

28. Who has the authority to accept or reject completed work at the end of a sprint?

- A) Scrum Master
- B) Developers
- C) Client's IT department
- D) Product Owner

29. According to the lecture, the Product Owner is NOT the same as which of the following?

- A) A stakeholder representative
- B) The actual client who sponsors the software
- C) The person who maintains the product backlog
- D) A member of the Scrum team

30. What is the “key focus” of the Scrum Master role as stated in the lecture materials?

- A) Maximizing the value of the product
- B) Delivering working software
- C) Helping the team work effectively
- D) Writing user stories

31. Which Scrum artifact contains a prioritized list of all requirements, features, and improvements for the product?

- A) Product Backlog
- B) Sprint Backlog
- C) Burn Down Chart
- D) Definition of Done

32. In the Product Backlog, where do the highest-value items typically appear?

- A) At the bottom
- B) In a random order
- C) Only in Release 3
- D) At the top

33. Why does the Product Backlog continue to change and evolve throughout a project?

- A) Because it is deleted and recreated every sprint
- B) Because customer feedback and new requirements are incorporated after each increment
- C) Because the Scrum Master edits it randomly
- D) Because it must match the Sprint Backlog exactly at all times

34. What is the Sprint Backlog best described as?

- A) A complete list of every requirement ever proposed for the product
- B) A chart showing remaining work versus time
- C) A subset of the Product Backlog containing items selected for a specific sprint
- D) A document only the client can edit

35. A Sprint Burn Down Chart typically plots which two variables?

- A) Work remaining vs. time
- B) Team salary vs. sprint number
- C) Number of bugs vs. number of developers

D) Customer satisfaction vs. cost

36. In a typical Sprint Burn Down Chart, which axis usually shows the outstanding work (backlog)?

- A) The horizontal axis
- B) Neither axis; it is shown as a pie chart
- C) It alternates each sprint
- D) The vertical axis

37. What is the standard template for writing a user story?

- A) As a , I will because
- B) As a , I want to so that
- C) As a , I verify using
- D) As a , I assign to

38. Which of the following is a well-written user story according to the lecture's criteria?

- A) As a customer, I want a database table to store account balances.
- B) As a member, I want an SQL query to search books.
- C) As a customer, I want to view my account balance so that I can monitor my finances.
- D) As a user, I want to generate reports.

39. Why is the user story “As a customer, I want a database table to store account balances” considered a poor example?

- A) It is too short
- B) It does not mention “As a”
- C) It includes a technical implementation detail (database) instead of focusing on user value
- D) It has too many benefits listed

40. Why is “As a user, I want to generate reports” considered an incomplete user story?

- A) It is missing the benefit (“so that...”) explaining why the user wants this
- B) It does not specify a technical database structure
- C) It uses too many words
- D) It does not name a specific report type

SECTION B — Multiple Select (Q41–60)

Select ALL correct options that apply (2 to 4 correct answers per question, out of options A–E).

41. Which of the following are challenges associated with traditional software development, as discussed in the lecture?

- A) Software is delivered in short, frequent increments
- B) Requirements may change during development
- C) Customer feedback is gathered and incorporated after every iteration
- D) Errors may be discovered late in the process
- E) Teams may build features that users no longer need

42. Which of the following are the 4 key values defined in the Agile Manifesto?

- A) Individuals and Interactions
- B) Comprehensive Documentation
- C) Customer Collaboration
- D) Contract Negotiation
- E) Responding to Change

43. Which of the following are recognized Agile development methodologies mentioned in the lecture?

- A) Scrum
- B) Kanban
- C) Waterfall
- D) Extreme Programming (XP)
- E) Spiral Model

44. Which of the following are considered Agile practices (as opposed to methodologies) in the lecture?

- A) Test-Driven Development (TDD)
- B) Scrum
- C) Pair Programming
- D) Behavior-Driven Development (BDD)
- E) Kanban

45. Which of the following are among the three main Scrum roles?

- A) Project Manager
- B) Business Analyst
- C) Product Owner
- D) Scrum Master
- E) Developers

46. Which of the following are responsibilities of the Product Owner?

- A) Facilitate daily scrum meetings and remove obstacles for the team
- B) Define and prioritize requirements
- C) Design, develop, and test the software code

- D) Maintain the Product Backlog
- E) Accept or reject completed work

47. Which of the following are responsibilities of the Scrum Master?

- A) Facilitate Scrum activities
- B) Promote collaboration and communication among the team
- C) Define and prioritize the product backlog
- D) Design and develop the software code
- E) Remove obstacles that affect the team's progress

48. Which of the following are the three main Scrum artifacts?

- A) Product Backlog
- B) Definition of Done
- C) Sprint Backlog
- D) Burn Down Chart
- E) Release Plan

49. Which of the following are among the four main Scrum activities discussed in the lecture?

- A) Sprint Planning
- B) Product Backlog Refinement
- C) Daily Scrum
- D) Sprint Review
- E) Sprint Retrospective

50. Which of the following statements about Sprint Planning are correct?

- A) It is conducted at the beginning of each Sprint
- B) Its purpose includes selecting Product Backlog items for the Sprint
- C) It results in a clear plan for the upcoming Sprint
- D) It is conducted after the Sprint Review to reflect on team performance
- E) It includes defining the Sprint Goal

51. Which of the following are true about the Daily Scrum?

- A) It typically lasts around 15 minutes
- B) It is held once per month
- C) It addresses what was completed, what's next, and any obstacles
- D) It is used to monitor daily progress
- E) It replaces the need for a Sprint Review

52. Which of the following are purposes of the Sprint Review?

- A) Demonstrate completed work
- B) Gather stakeholder feedback
- C) Reflect only on team performance with no stakeholder involvement
- D) Review progress toward the Sprint Goal
- E) Update the Product Backlog if needed

53. Which of the following are true about the Sprint Retrospective?

- A) It focuses on the team and the process
- B) It is conducted after the Sprint Review
- C) Its main purpose is to demonstrate the product to the client
- D) It aims to identify what went well and areas for improvement
- E) It results in a plan for improvements in the next Sprint

54. Which of the following are elements that should be included in a well-written user story?

- A) A clearly identified user role
- B) Technical implementation details like database structures
- C) A description of the goal the user wants to achieve
- D) An explanation of the benefit ("so that...")
- E) SQL queries or code snippets describing functionality

55. Based on the burndown chart activity (120 tasks over 5 days: Day1=20, Day2=50, Day3=0, Day4=20, Day5=30), which of the following statements are correct?

- A) The ideal burn down velocity is 24 tasks per day
- B) Day 2 had the highest productivity with 50 tasks completed
- C) Day 3 had the lowest productivity with 0 tasks completed
- D) The team completed all 120 tasks by the end of Day 5
- E) The team completed only 100 tasks in total

56. Which of the following correctly match an Agile value to what it replaces in the Traditional way?

- A) Individuals and Interactions over Process and Tools
- B) Working Software over Comprehensive Documentation
- C) Customer Collaboration over Contract Negotiation
- D) Responding to Change over Comprehensive Documentation
- E) Working Software over Following a Plan

57. In the Hospital Appointment Management System example, which of the following user stories were assigned HIGH priority in the initial Product Backlog?

- A) As a patient, I want to register an account so that I can use the appointment system.
- B) As a patient, I want to receive appointment reminders so that I do not miss my appointment.
- C) As a patient, I want to search for doctors by specialization so that I can find the correct doctor.
- D) As a doctor, I want to view my daily appointments so that I can prepare for consultations.
- E) As a patient, I want to book an appointment so that I can reserve a consultation time.

58. Which of the following are valid reasons given in the lecture for why Sprint 1 of the Hospital Appointment System focused on registration and doctor search?

- A) They have the highest business value
- B) They are core system functionality needed for the system to operate
- C) They are the lowest-priority items in the backlog
- D) They form a foundation for future features
- E) They allow the team to deliver a working product early

59. Which of the following statements accurately describe the difference between Agile "methodologies" and "practices"?

- A) Methodologies provide the overall structure for managing a project
- B) Practices are specific techniques used during implementation
- C) TDD and Pair Programming are examples of methodologies
- D) Scrum and Kanban are examples of methodologies
- E) Practices provide the overall project structure while methodologies focus on implementation

60. Which of the following are true regarding the difference between a Client and a Product Owner, as clarified in the lecture?

- A) The Product Owner is the same person as the Client
- B) The Client sponsors the software and communicates the initial requirements
- C) The Product Owner represents customers and stakeholders within the Scrum team
- D) The Product Owner prioritizes the requirements after they are collected
- E) The Client is always a member of the development team

ANSWER KEY & EXPLANATIONS

Section A — Single-Answer Multiple Choice

Q1: B

Traditional models struggle with late changes because significant work is already complete, leading to rework, higher cost, and delays.

Q2: D

Agile was introduced specifically to handle changing requirements more effectively than traditional models.

Q3: A

The Traditional Way is described as high cost, resistant to change, and lacking transparency.

Q4: C

The Modern Way delivers the highest value and quality within the shortest time, unlike the Traditional Way.

Q5: B

This is the exact example used in the lecture to show why late changes are costly and difficult in traditional development.

Q6: D

Delivering software in small, frequent increments is how Agile helps, not a traditional challenge.

Q7: A

Agile helps by delivering in increments and adapting to changing requirements as they arise.

Q8: C

The lecturer states the Agile Manifesto came into the industry around 2000-2001.

Q9: B

Values define the philosophy (e.g., wanting to be healthy); principles are the actions that apply that philosophy in practice.

Q10: D

The Agile Manifesto defines 4 key Values and 12 Principles.

Q11: A

Individuals and Interactions emphasizes team collaboration and customer involvement over rigid processes and tools.

Q12: C

Working Software is valued over Comprehensive Documentation in the Agile Manifesto.

Q13: B

Responding to Change is valued over Following a Plan.

Q14: D

Scrum organizes work into short iterations called Sprints, typically one to four weeks.

Q15: A

Kanban uses a visual board (with columns and sticky notes/cards) to track the progress of work.

Q16: C

XP focuses on engineering practices and ensuring high code quality.

Q17: B

Lean Software Development focuses on eliminating waste and maximizing the value delivered.

Q18: D

Sprints usually last one to four weeks, commonly two to three weeks per the lecturer.

Q19: A

Methodologies (e.g., Scrum, Kanban) give overall project structure; practices (e.g., TDD, Pair Programming) are specific implementation techniques.

Q20: C

In TDD, test cases are written first, and code is then written to satisfy those tests.

Q21: B

Pair programming lets two developers catch bugs and improve quality collaboratively and quickly.

Q22: D

BDD defines system behavior using business-focused scenarios understandable by both business and technical people.

Q23: A

Scrum is an Agile framework that divides work into Sprints and relies on team collaboration.

Q24: D

The Product Owner represents the customer/stakeholders and prioritizes requirements.

Q25: B

The Scrum Master acts as a service leader/facilitator, not a task-assigning manager.

Q26: C

The lecture explicitly clarifies the Scrum Master is not a traditional project manager or team lead.

Q27: A

Developers design, build, test, and deliver the working product increment each sprint.

Q28: D

The Product Owner accepts or rejects completed work based on whether it meets requirements.

Q29: B

The lecture stresses the Product Owner is not the actual client; the client sponsors the software, while the PO represents customers/stakeholders.

Q30: C

The Scrum Master's key focus is helping the team work effectively.

Q31: A

The Product Backlog is the prioritized list of all requirements, features, and improvements.

Q32: D

High-value items appear at the top of the Product Backlog; lower-valued items toward the bottom.

Q33: B

The Product Backlog evolves as customer feedback and new requirements are added after each increment.

Q34: C

The Sprint Backlog is a subset of the Product Backlog for the items selected for a particular sprint.

Q35: A

A burn down chart is a graphical representation of work remaining versus time.

Q36: D

The outstanding work/backlog is usually on the vertical axis, with time along the horizontal axis.

Q37: B

The standard user story template is: As a , I want to so that .

Q38: C

This option correctly identifies a user role, a goal, and a benefit, without technical jargon.

Q39: C

User stories should avoid technical implementation details like databases and focus on user value.

Q40: A

The story lacks the “so that...” clause explaining the benefit to the user.

Section B — Multiple Select

Q41: B, D, E

Frequent increments and continuous feedback describe Agile, not traditional challenges.

Q42: A, C, E

Comprehensive Documentation and Contract Negotiation are the traditional counterparts, not Agile values.

Q43: A, B, D

Waterfall and Spiral are traditional software process models, not Agile methodologies.

Q44: A, C, D

Scrum and Kanban are methodologies (overall structure), while TDD, Pair Programming, and BDD are implementation practices.

Q45: C, D, E

Project Manager and Business Analyst are not defined Scrum roles; the three Scrum roles are Product Owner, Scrum Master, and Developers.

Q46: B, D, E

Facilitating scrum activities is the Scrum Master's job; designing/coding is the Developers' job.

Q47: A, B, E

Prioritizing the backlog is the Product Owner's job; coding is the Developers' job.

Q48: A, C, D

The lecture names Product Backlog, Sprint Backlog, and Burn Down Chart as the three main Scrum artifacts.

Q49: A, C, D, E

The lecture lists Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective as the four main Scrum activities.

Q50: A, B, C, E

Reflecting on team performance after the Sprint Review describes the Sprint Retrospective, not Sprint Planning.

Q51: A, C, D

The Daily Scrum is held daily (not monthly) and does not replace the Sprint Review.

Q52: A, B, D, E

Reflecting only on team performance without stakeholders describes the Sprint Retrospective, not the Sprint Review.

Q53: A, B, D, E

Demonstrating the product to the client is the purpose of the Sprint Review, not the Retrospective.

Q54: A, C, D

Good user stories avoid technical implementation details such as databases or SQL queries.

Q55: A, B, C, D

120 tasks / 5 days = 24 tasks/day ideal velocity. $20+50+0+20+30 = 120$ tasks total, so the team completed all tasks.

Q56: A, B, C

Responding to Change is valued over Following a Plan (not Comprehensive Documentation), and Working Software is valued over Comprehensive Documentation (not Following a Plan) — D and E swap the pairs incorrectly.

Q57: A, C, E

Registration, doctor search, and appointment booking were High priority; reminders were Low and the doctor dashboard was Medium priority.

Q58: A, B, D, E

These items were selected because they are high-value, core, foundational, and allow early delivery — not because they were low priority.

Q59: A, B, D

TDD and Pair Programming are practices, not methodologies, and option E reverses the correct definitions.

Q60: B, C, D

The lecture explicitly states the Product Owner is not the same person as the client.